# Prime Time

Department of Computer Science and Engineering
National Sun Yat-sen University

TA: 丁祥益

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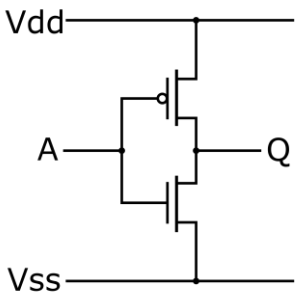
# Outline

1. Introduction
2. Goal
3. Manual

# 1. Introduction

```
                                        Int       Switch    Leak      Total
Hierarchy                               Power     Power     Power     Power     %
-----------------------------------------------------------------------------------
```

- Total Power = Leakage Power + Dynamic Power
  - **Leakage(static) Power** : Power consumed by sub-threshold and gate leakage currents within transistors, even when they are in a static "off" state.
  - **Dynamic Power**
    - Switching Power : Power consumed by the charging and discharging of the external load capacitance only when the cell's output signal transitions.
    - **Internal Power**
      - Short-Circuit Power : Transient current flowing from VDD to GND when both PMOS and NMOS networks are momentarily "on".
      - **Internal Capacitance** : Power from charging/discharging parasitic capacitances on internal nodes within the cell.



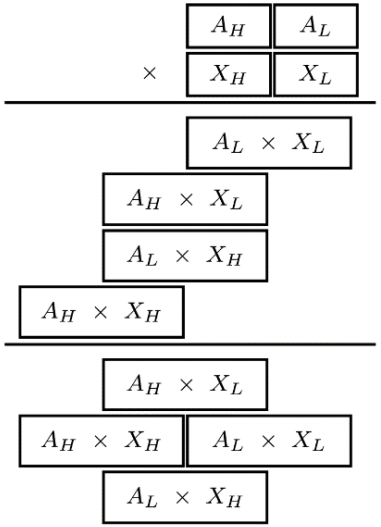
# 2. Goal --- average power report

- without clock gate

| Hierarchy | Int Power | Switch Power | Leak Power | Total Power | % |
|---|---|---|---|---|---|
| top | 6.08e-04 | 4.38e-04 | 2.28e-04 | 1.27e-03 | 100.0 |
| a1_h (adder_2) | 2.65e-05 | 6.40e-05 | 1.63e-05 | 1.07e-04 | 8.4 |
| r328 (adder_2_DW01_add_0) | 5.93e-06 | 5.53e-06 | 2.38e-06 | 1.38e-05 | 1.1 |
| r329 (adder_2_DW01_sub_0) | 6.53e-06 | 6.60e-06 | 2.89e-06 | 1.60e-05 | 1.3 |
| r330 (adder_2_DW_cmp_0) | 5.26e-07 | 2.83e-06 | 4.73e-07 | 3.82e-06 | 0.3 |
| a2_h (adder_1) | 4.82e-06 | 5.34e-06 | 2.81e-06 | 1.30e-05 | 1.0 |
| r328 (adder_1_DW01_add_0) | 4.36e-06 | 3.02e-06 | 2.14e-06 | 9.52e-06 | 0.7 |
| r329 (adder_1_DW01_sub_0) | 5.93e-06 | 5.53e-06 | 2.38e-06 | 1.38e-05 | 1.1 |
| r330 (adder_1_DW_cmp_0) | 4.36e-06 | 3.02e-06 | 2.14e-06 | 9.52e-06 | 0.7 |
| a3_h (adder1_1) | 9.66e-07 | 2.78e-08 | 5.16e-08 | 1.05e-06 | 0.1 |
| r328 (adder1_1_DW01_add_0) | 4.82e-06 | 5.34e-06 | 2.81e-06 | 1.30e-05 | 1.0 |
| r329 (adder1_1_DW01_sub_0) | 4.36e-06 | 3.02e-06 | 2.14e-06 | 9.52e-06 | 0.7 |
| r330 (adder1_1_DW_cmp_0) | 6.87e-06 | 6.51e-06 | 2.12e-06 | 1.55e-05 | 1.2 |
| p0 (pipeline0_1) | 1.32e-04 | 1.50e-05 | 6.91e-06 | 1.54e-04 | 12.1 |
| p2_1_h (pipeline2_1) | 3.11e-05 | 9.22e-07 | 2.14e-06 | 3.42e-05 | 2.7 |
| p1 (pipeline1_0) | 2.98e-05 | 3.30e-06 | 1.65e-06 | 3.48e-05 | 2.7 |
| p2 (pipeline1_7) | 3.02e-05 | 2.96e-06 | 1.61e-06 | 3.48e-05 | 2.7 |
| p3 (pipeline1_6) | 2.97e-05 | 3.01e-06 | 1.57e-06 | 3.42e-05 | 2.7 |
| p4 (pipeline1_5) | 3.00e-05 | 3.61e-06 | 1.68e-06 | 3.53e-05 | 2.8 |
| p_h (pipeline0_0) | 3.37e-06 | 3.77e-06 | 7.96e-06 | 1.51e-05 | 1.2 |
| clk_gate_C301 (SNPS_CLOCK_GATE_HIGH_top) | | | | | |
| | 6.64e-07 | 0.000 | 4.91e-08 | 7.13e-07 | 0.1 |
| p2_1 (pipeline2_0) | 3.66e-05 | 4.67e-06 | 1.93e-06 | 4.32e-05 | 3.4 |
| p0m (pipelinem_0) | 9.66e-07 | 2.78e-08 | 5.16e-08 | 1.05e-06 | 0.1 |
| n32 (normal32) | 1.58e-06 | 6.57e-06 | 5.76e-06 | 1.39e-05 | 1.1 |
| add_32 (normal32_DW01_add_2) | 0.000 | 0.000 | 1.42e-06 | 1.42e-06 | 0.1 |
| a1 (adder_0) | 2.65e-05 | 6.40e-05 | 1.63e-05 | 1.07e-04 | 8.4 |
| r328 (adder_0_DW01_add_0) | 5.93e-06 | 5.53e-06 | 2.38e-06 | 1.38e-05 | 1.1 |
| r329 (adder_0_DW01_sub_0) | 6.53e-06 | 6.60e-06 | 2.89e-06 | 1.60e-05 | 1.3 |

- with clock gate

| Hierarchy | Int Power | Switch Power | Leak Power | Total Power | % |
|---|---|---|---|---|---|
| top | 5.21e-04 | 5.46e-05 | 2.17e-04 | 7.92e-04 | 100.0 |
| a1_h (adder_2) | 0.000 | 0.000 | 1.41e-05 | 1.41e-05 | 1.8 |
| r328 (adder_2_DW01_add_0) | 0.000 | 0.000 | 2.53e-06 | 2.53e-06 | 0.3 |
| r329 (adder_2_DW01_sub_0) | 0.000 | 0.000 | 2.97e-06 | 2.97e-06 | 0.4 |
| r330 (adder_2_DW_cmp_0) | 0.000 | 0.000 | 5.35e-07 | 5.35e-07 | 0.1 |
| a2_h (adder_1) | 0.000 | 0.000 | 1.51e-05 | 1.51e-05 | 1.9 |
| r328 (adder_1_DW01_add_0) | 0.000 | 0.000 | 2.12e-06 | 2.12e-06 | 0.3 |
| r329 (adder_1_DW01_sub_0) | 0.000 | 0.000 | 2.78e-06 | 2.78e-06 | 0.4 |
| r330 (adder_1_DW_cmp_0) | 0.000 | 0.000 | 5.35e-07 | 5.35e-07 | 0.1 |
| a3_h (adder1_1) | 0.000 | 0.000 | 1.52e-05 | 1.52e-05 | 1.9 |
| r328 (adder1_1_DW01_add_0) | 0.000 | 0.000 | 2.14e-06 | 2.14e-06 | 0.3 |
| r329 (adder1_1_DW01_sub_0) | 0.000 | 0.000 | 3.36e-06 | 3.36e-06 | 0.4 |
| r330 (adder1_1_DW_cmp_0) | 0.000 | 0.000 | 7.43e-07 | 7.43e-07 | 0.1 |
| p0 (pipeline0_1) | 1.22e-04 | 5.02e-06 | 6.34e-06 | 1.33e-04 | 16.8 |
| p2_1_h (pipeline2_1) | 3.01e-05 | 1.23e-06 | 1.76e-06 | 3.31e-05 | 4.2 |
| p1 (pipeline1_0) | 2.64e-05 | 1.04e-06 | 1.49e-06 | 2.89e-05 | 3.7 |
| p2 (pipeline1_7) | 2.62e-05 | 1.04e-06 | 1.48e-06 | 2.87e-05 | 3.6 |
| p3 (pipeline1_6) | 2.59e-05 | 1.04e-06 | 1.42e-06 | 2.84e-05 | 3.6 |
| p4 (pipeline1_5) | 2.66e-05 | 1.04e-06 | 1.53e-06 | 2.92e-05 | 3.7 |
| p_h (pipeline0_0) | 1.22e-04 | 5.02e-06 | 6.35e-06 | 1.33e-04 | 16.8 |
| clk_gate_C301 (SNPS_CLOCK_GATE_HIGH_top) | | | | | |
| | 9.44e-07 | 0.000 | 5.23e-08 | 9.96e-07 | 0.1 |
| p2_1 (pipeline2_0) | 3.03e-05 | 1.23e-06 | 1.76e-06 | 3.33e-05 | 4.2 |
| p0m (pipelinem_0) | 8.49e-07 | 3.70e-08 | 5.18e-08 | 9.38e-07 | 0.1 |
| n32 (normal32) | 3.01e-06 | 1.80e-05 | 5.81e-06 | 2.68e-05 | 3.4 |
| add_32 (normal32_DW01_add_2) | 4.49e-07 | 1.48e-06 | 1.43e-06 | 3.36e-06 | 0.4 |
| a1 (adder_0) | 0.000 | 0.000 | 1.53e-05 | 1.53e-05 | 1.9 |



# 3. Manual (1/2)

**Design Compiler**
Needed:
PE.v
TsmcXX.v

Dump:
PE_gate.v
PE_syn.sdc
PE_syn.sdf
PE_syn.ddc

→

**Post simulation**
Needed:
PE_gate.v
testbenth.v
PE_syn.sdf

Dump:
PE_wave.vcd

→

**Primetime**
Needed:
PE_gate.v
PE_syn.sdc
PE_syn.sdf
PE_wave.vcd

Dump:
Report_power_average_vcd.rpt

gate_level_TB.v

```
// prime time
initial begin
    $dumpvars();
    $dumpfile("DP_4_wave.vcd");

    $sdf_annotate("/home/m133040045/ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.sdf", test_module);
end
```

- verilog (gate level netlist)
- ddc (save your design)
- sdc (Synopsys Design Constraints)
- sdf (Standard Delay Format)
- vcd (waveform file)

# 3. Manual (2/2)

```
#####################################################################
# power analysis
#####################################################################
# source saifmap.ptpx.tcl
# report_name_mapping
set Top DP_4
set power_enable_analysis true
set power_analysis_mode averaged
set power_report_leakage_breakdowns true
# gate-level .v
read_verilog /home/m133040045/back_up/m133040045/ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.v
#rtl top module name
current_design $Top
link

read_sdc /home/m133040045/back_up/m133040045/ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.sdc
read_sdf /home/m133040045/back_up/m133040045/ALU/HW3_ALU/FLP_DP_4/gate/delay/DP_4_syn.sdf
check_timing
update_timing

read_vcd -strip_path testbench/test_module /home/m133040045/back_up/m133040045/ALU/HW3_ALU/FLP_DP_4/postsim/DP_4_wave.vcd

check_power
update_power

report_power -hierarchy > report_${Top}_power_average_vcd.rpt

#quit
exit
```

top module name of your design

file path of gate-level netlist

```
module testbench();
    integer file_32, file_16, file_ans32, file_ans16;

    reg CLK = 0;
    reg RST = 0;
    reg precision = 0;
    reg     [31:0] data32[0:`DATA_NUM-1];
    reg     [31:0] data16[0:`DATA_NUM-1];
    reg     [63:0] temp_answer;
    reg     [10:0] exp;
    reg     [31:0] input_a1, input_a2, input_a3, input_a4, input_b1, input_b2, inp
    wire     [31:0] answer;
    reg     [31:0]  answer_shift[0:`PIPE];  // answer shift reg
    reg            MODE;
    wire    [31:0] outcome;
    reg stop;
    real real_a1, real_a2, real_a3, real_a4, real_b1, real_b2, real_b3, real_b4, r

    DP_4 test_module( .clk(CLK), .rst(RST), .precision(precision), .a1(input_a1),
                      .c(outcome));
```

gate_level_TB.v

**END**